

Explainable Prediction of Mutagenicity Using SHAP

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Abstract. Predicting whether chemical compounds are mutagenic is important for drug discovery and toxicology. However, machine learning models that make these predictions are often black boxes, making it hard to trust their decisions. In this project, we trained a Gradient Boosting model on the MUTAG dataset to predict mutagenicity and used SHAP (SHapley Additive exPlanations) to explain the model’s predictions. Our model achieved 75% test accuracy. SHAP analysis showed that molecular size and structural features like 5-membered rings and reactive functional groups are the most important predictors. We evaluated our explanations using fidelity checks and feature removal tests, confirming that SHAP provides reliable interpretations of the model’s behavior.

Keywords: Mutagenicity · SHAP · Explainable AI · Gradient Boosting

1 Introduction

Understanding why a machine learning model makes a prediction is important in safety-critical domains like toxicology. If a model incorrectly labels a harmful compound as safe, the consequences can be serious. Traditional black-box models give good accuracy but don’t tell us why they make decisions. Explainable AI (XAI) methods help us understand these decisions.

The goal of this project is to:

1. Build a model that predicts if a chemical compound is mutagenic.
2. Explain the model’s predictions using SHAP.
3. Check if the explanations are reliable.

We used SHAP because it tells us how much each feature contributes to a prediction. It works for both global understanding (which features matter most overall) and local understanding (why a specific compound was predicted a certain way).

We used the MUTAG dataset, which contains chemical compounds labeled as mutagenic or non-mutagenic. The data is stored as an RDF graph with over 74,000 triples representing atoms, bonds, and test results.

We followed Strategy 2 from the course: convert graph data to a table, train a classical ML model, and then explain it with SHAP.

2 Data Analysis

2.1 Dataset Overview

The MUTAG dataset has 340 compounds labeled as either mutagenic or non-mutagenic. We extracted the following statistics from the RDF graph:

Table 1. Dataset Summary Statistics

Metric	Value
Total Triples	74,227
Total Unique Nodes	23,644
URI Nodes	22,540
Literal Nodes	1,104
Unique Predicates	23
Unique Node Types	113
Total Labeled Compounds	340
Training Samples	272
Test Samples	68
Mutagenic (label=1)	129
Non-Mutagenic (label=0)	211

2.2 Feature Engineering

We converted the RDF graph to a table where each row represents a compound and each column represents a property. We created three types of features:

- **Structural fragments:** Extracted from the `hasStructure` property (38 types, one-hot encoded).
- **Count features:** Number of atoms and bonds per compound.
- **Biological test results:** One-hot encoded from columns like `salmonella`, `cytogen_ca`, and `amesTestPositive`.

The final feature matrix had 340 samples and 77 features (reduced to 76 after removing zero-variance features).

2.3 Class Distribution

The dataset has more non-mutagenic compounds than mutagenic ones:

- Non-mutagenic (0): 211 compounds
- Mutagenic (1): 129 compounds

This imbalance matters when evaluating model performance, as the model might favor predicting the majority class.

3 Model Training & Evaluation

3.1 Models Used

We trained three models and compared their performance:

Table 2. Model Performance Comparison

Model	Train Acc	Test Acc	Precision	Recall	F1
Gradient Boosting	99.3%	75.0%	0.67	0.52	0.59
Random Forest	100.0%	73.5%	0.61	0.48	0.54
Decision Tree	84.2%	67.6%	0.53	0.39	0.45

3.2 Key Findings

1. **Overfitting:** All models show a gap between training and test accuracy. This is expected given the small dataset (272 training samples) and high number of features.
2. **Gradient Boosting performs best:** It achieved the highest test accuracy (75.0%) and F1 score (0.59), so we selected it for SHAP explanation.
3. **Low recall for mutagenic class:** Recall ranges from 0.39 to 0.52. This means the model misses about half of the truly mutagenic compounds. False negatives are more dangerous in safety-critical applications.

3.3 Confusion Matrix

For the Gradient Boosting model:

Table 3. Confusion Matrix - Gradient Boosting

	Predicted Safe	Predicted Mutagenic
Actual Safe	39	6
Actual Mutagenic	11	12

The model correctly identifies 87% of non-mutagenic compounds but only 52% of mutagenic ones.

4 Model Explanation

4.1 Global Explanations

We used SHAP to understand which features are most important overall.

Bar Plot: Feature Importance The bar plot shows features ranked by mean absolute SHAP value. Higher bars mean the feature has more influence on predictions.

Table 4. Top 15 Most Important Features

Rank	Feature	Mean SHAP	Category
1	num_bonds	0.70	Count
2	num_atoms	0.65	Count
3	struct_five_ring	0.61	Structural
4	cytogen_ca_false	0.38	Biological
5	struct_amine	0.30	Structural
6	struct_methyl	0.29	Structural
7	struct_alkyl_halide	0.26	Structural
8	cytogen_ca_true	0.25	Biological
9	struct_ester	0.18	Structural
10	salmonella_false	0.16	Biological
11	struct_non_ar_hetero_6_ring	0.16	Structural
12	struct_non_ar_6c_ring	0.16	Structural
13	struct_methoxy	0.15	Structural
14	struct_six_ring	0.14	Structural
15	cytogen_sce_missing	0.13	Biological

Key insights:

- **Molecular size matters most:** num_bonds and num_atoms are the top two features. Larger molecules with more atoms and bonds are more likely to be mutagenic.
- **Structural features are important:** Five-membered rings rank third, followed by amines, methyl groups, and alkyl halides. These are known reactive groups that can interact with DNA.
- **Biological tests provide supporting evidence:** Negative test results are more important than positive ones, as they give strong evidence of safety.

By category:

- Structural features: 2.26 total importance
- Count features: 1.35 total importance
- Biological features: 0.91 total importance

Beeswarm Plot: Feature Direction The beeswarm plot shows both importance and direction. Red dots mean high feature values, blue dots mean low feature values. The right side pushes toward mutagenic, the left side pushes toward safe.

Key observations:

- **Bigger molecules = more mutagenic:** Red dots (big molecules) are on the right.
- **Five-membered rings:** Red dots are on the right, meaning presence pushes toward mutagenic.
- **Reactive groups:** Amines, methyl groups, and alkyl halides push toward mutagenic.

4.2 Local Explanations

Mutagenic Compound: Starting from the baseline prediction (-0.81), this compound’s score rose to +2.65, making it highly likely to be mutagenic.

Features pushing toward mutagenic:

- `struct_alkyl_halide` (+0.98): Presence of an alkyl halide group
- `num_bonds` (+0.68): Many bonds in the molecule
- `struct_non_ar_6c_ring` (+0.57): Non-aromatic ring
- `cytogen_ca_false` (+0.57): Negative test result
- `cytogen_ca_true` (+0.56): Positive test result
- `struct_amine` (+0.45): Amine group

This compound has many features that push toward mutagenic, including size, reactive groups, and mixed test results.

Non-Mutagenic Compound: Starting from the baseline prediction (-0.81), this compound’s score dropped to -2.3, making it highly likely to be non-mutagenic.

Features pushing toward safe:

- `cytogen_ca_true` (-0.59): Positive test result
- `struct_sulfide` (-0.45): Sulfide group
- `struct_ethyl` (-0.35): Ethyl group
- `salmonella_n_false` (-0.25): Negative test

Features pushing toward mutagenic:

- `struct_amine` (+0.55): Amine group

Even though this compound has an amine group, most features push toward safe, resulting in a non-mutagenic prediction.

4.3 Evaluation of Explanations

Fidelity Check We checked if SHAP values faithfully reproduce the model’s predictions. The correlation between SHAP reconstruction and model scores was 1.0, meaning the explanations are fully faithful to the model.

Feature Removal Test We tested if SHAP-identified features are genuinely important by removing top SHAP features and random features:

Table 5. Feature Removal Test Results

k	Baseline	SHAP Removal	Random Removal	SHAP Drop	Random Drop
5	0.75	0.82	0.75	-0.07	0.00
10	0.75	0.79	0.79	-0.04	-0.04
15	0.75	0.79	0.76	-0.04	-0.01
20	0.75	0.76	0.74	-0.01	0.01

The feature removal test results were mixed. At $k=5$, removing SHAP features actually improved accuracy slightly (-0.0735 vs 0.0000). At $k=10$, both SHAP and random removal had identical drops (-0.0441). At $k=15$ and 20, SHAP removal caused slightly larger drops ($k=15$: -0.0441 vs -0.0147; $k=20$: -0.0147 vs +0.0147). The differences are small and not conclusive.

This suggests that on small datasets like ours, feature removal tests can be noisy. The perfect fidelity score (1.0) provides much stronger evidence that SHAP explanations are reliable and faithfully represent the model’s decisions.

5 Conclusion

We trained a Gradient Boosting model to predict mutagenicity on the MUTAG dataset and used SHAP to explain its predictions. The model achieved 75% test accuracy, though recall for the mutagenic class (0.52) remains a limitation.

SHAP analysis revealed that molecular size and structural features are the most important predictors. Local explanations showed how specific features push compounds toward mutagenic or safe predictions.

We evaluated our explanations using fidelity checks and feature removal tests. The perfect fidelity score (1.0) confirms SHAP values faithfully reconstruct the model's decisions. The feature removal test yielded mixed results but showed some evidence that SHAP identifies genuinely important features.

Our results show that SHAP is a useful tool for explaining black-box models in toxicology, helping domain experts understand and trust model predictions. Future work could address class imbalance and explore other explanation methods.

6 Contributions

Maaz Adnan: Data processing, Model training, SHAP analysis, evaluation, report writing.

Muhammad Ali: Data analysis, SHAP analysis, interpretation, report writing.

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References

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